Isolated magnetic moments

Atom in magnetic field: $H = \sum_{i=1}^{Z} \frac{(\vec{p}_i + e\vec{A}(\vec{r}_i))^2}{2m_e} + g\mu_B \vec{B}\cdot\sum_{i=1}^{Z}\vec{S}_i + V_i$

(Z electrons)

P.E. of i-th electron

$$\vec{B} = \vec{\nabla}\times\vec{A} \quad , \quad \vec{A}(\vec{r}) = \frac{\vec{B}\times\vec{r}}{2} = \frac{1}{2}B(x\hat{y} - y\hat{x})$$

for $\vec{B} = B\hat{z}$

Then $\vec{p}\cdot\vec{A} = -i\hbar\vec{\nabla}\cdot\frac{1}{2}B(x\hat{y} - y\hat{x}) = -\frac{i\hbar B}{2}\left(\frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x)\right) = 0$

$$\Rightarrow (\vec{p} + e\vec{A})^2 = \vec{p}^{\,2} + e^2\vec{A}^{\,2} + 2e\vec{A}\cdot\vec{p}$$

$$H = \sum_i \left\{\frac{\vec{p}_i^{\,2}}{2m} + V_i + \frac{e}{2m_e}\vec{B}\times\vec{r}_i\cdot\vec{p}_i + \frac{e^2}{8m_e}(\vec{B}\times\vec{r}_i)^2 + g\mu_B\vec{B}\cdot\vec{S}_i\right\} = \sum_i\left(\frac{\vec{p}_i^{\,2}}{2m_e} + V_i\right) +$$

$$+ \underbrace{\frac{\mu_B}{\hbar}(\vec{L} + g\vec{S})\cdot\vec{B}}_{\text{paramagnetic term}} + \underbrace{\frac{e^2}{8m_e}\sum_{i=1}^{Z}(\vec{B}\times\vec{r}_i)^2}_{\text{diamagnetic term}}$$

$$\vec{L} = \sum_i \vec{r}_i\times\vec{p}_i = \sum_i \vec{\ell}_i$$

$$\vec{B}\times\vec{r}\cdot\vec{p} = \varepsilon_{abc}B_a x_b p_c = -\varepsilon_{bac}B_a x_b p_c =$$
$$= \varepsilon_{bca}x_b p_c B_a = \vec{r}\times\vec{p}\cdot\vec{B}$$

Will see later that $\vec{L} + g\vec{S} = g_J\vec{J}$: this replacement is understood to apply to matrix elements.

Diamagnetism

Suppose atom with <u>filled shells</u> $\Rightarrow \langle \vec{L} \rangle = 0$, $\langle \vec{S} \rangle = 0$, $\langle \vec{J} \rangle = 0$

where $|\rangle \equiv |0\rangle$ is ~~the~~ atomic ground state.

Then $H = H_0 + \frac{e^2}{8m} \sum_{i=1}^{Z} (\vec{B} \times \vec{r}_i)^2$

let $\vec{B} = (0, 0, B) \Rightarrow \vec{B} \times \vec{r}_i = \varepsilon^{\alpha\beta\gamma} B^\beta r_i^\gamma = (\beta = z) =$

$$= -B y_i \hat{x} + B x_i \hat{y}$$

$$\Rightarrow (\vec{B} \times \vec{r}_i)^2 = B^2 (x_i^2 + y_i^2)$$

$\Rightarrow$ correction to the ground state energy

$$\Delta E_0^{(1)} = \frac{e^2 B^2}{8m} \sum_{i=1}^{Z} \langle 0 | x_i^2 + y_i^2 | 0 \rangle = \frac{e^2 B^2}{12m} \sum_{i=1}^{Z} \langle 0 | r_i^2 | 0 \rangle$$

For spherically symmetric atom, $\langle x_i^2 \rangle = \langle y_i^2 \rangle = \frac{1}{3} \langle \vec{r}_i^{\,2} \rangle$

$$M = -\left(\frac{\partial F}{\partial B}\right)_{T,V} = -\frac{N}{V} \frac{\partial \Delta E_0^{(1)}}{\partial B} = -\frac{N e^2 B}{6 m V} \sum_{i=1}^{Z} \langle r_i^2 \rangle$$

at $T=0$

(N atoms per volume V) $\frac{N}{V} = n$ = concentration

~~At~~ susceptibility $\chi_{dia} = \frac{M}{H} = \frac{\mu_0 M}{B} = -\frac{N e^2 \mu_0}{6 m V} \sum_{i=1}^{Z} \langle r_i^2 \rangle$

Note that $\chi_{dia} < 0$

water $\chi = -90 \cdot 10^{-6}$

NaCl $-13.9 \cdot 10^{-6}$

~~graphite~~ benzene $-7.2 \cdot 10^{-6}$

<u>Van Vleck susceptibility</u> $\vec{J} = 0$ but $\vec{L}, \vec{S} \neq 0$

Even though $\langle 0 | (\vec{L} + g\vec{S}) \cdot \vec{B} | 0 \rangle = 0$, there should be higher order contribution.

In second order $\Delta E_0^{(2)} = -\sum_{n>0} \mu_B^2 \frac{|\langle 0 | (\vec{L} + g\vec{S}) \cdot \vec{B} | n \rangle|^2}{E_n - E_0} =$

$$= -\sum_{n>0} \mu_B^2 B^2 \frac{|\langle 0 | (L_z + g S_z) | n \rangle|^2}{E_n - E_0}$$

$$\Rightarrow \Delta E_0 = \Delta E_0^{(1)} + \Delta E_0^{(2)} = B^2 \left(\frac{e^2}{12m} \sum_{i=1}^{Z} \langle 0 | r_i^2 | 0 \rangle - \right.$$

$$\left. - \sum_{n>0} \mu_B^2 \frac{|\langle 0 | (L_z + g S_z) | n \rangle|^2}{E_n - E_0} \right)$$

$$\Rightarrow M = - n \frac{\partial \Delta E_0}{\partial B} = nB \left(\overbrace{2\mu_B^2 \sum_{n>0} \frac{|\langle 0 | (L_z + g S_z) | n \rangle|^2}{E_n - E_0}}^{\text{Van Vleck paramagnetism}} \right.$$

$$\left. - \frac{e^2}{6m} \sum_{i=1}^{Z} \langle 0 | r_i^2 | 0 \rangle \right).$$

Paramagnetism

Atom with finite magnetic moment $\vec{\mu} = \vec{\mu}_L + \vec{\mu}_S$ (unfilled shell).

$$H_Z = - \vec{\mu} \cdot \vec{B}$$

$$\vec{\mu} = \frac{\mu_B}{\hbar} \left(\vec{L} + 2\vec{S} \right)$$

Generally, $\vec{L}$ and $\vec{S}$ are not conserved separately, but $\vec{J} = \vec{L} + \vec{S}$ is!

Must be that $\vec{\mu} = \frac{\mu_B}{\hbar} g_J \vec{J}$ where g_J = Lande g-factor

$$g_J = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}$$

$$H_Z = - \frac{\mu_B}{\hbar} g_J \vec{J} \cdot \vec{B} = - \frac{\mu_B}{\hbar} g_J B \hat{J}_z$$

but $\hat{J}_z |m\rangle = \hbar m_J |m\rangle$, with $m_J = \underbrace{-J, -J+1 \ldots J-1, J}_{2J+1}$

$$\Rightarrow H_Z = - \mu_B g_J B m_J$$

Partition function

$$Z = \sum_{m=-J}^{J} e^{\frac{\mu_B g_J B}{k_B T} m} = \sum_{m=-J}^{J} e^{xm} \quad , \; x \equiv \frac{\mu_B g_J B}{k_B T}$$

$$\langle m \rangle = \frac{\sum m e^{xm}}{\sum e^{xm}} = \frac{1}{Z} \frac{\partial Z}{\partial x}$$

Magnetic moment $\quad M = \frac{N}{V} g_J \mu_B \langle m \rangle = \frac{n g_J \mu_B}{Z} \frac{\partial Z}{\partial B} \frac{\partial B}{\partial x} =$

$$= n k_B T \frac{\partial \ln Z}{\partial B}$$

But $\quad Z = \sum_{m=-J}^{J} e^{xm} = e^{-Jx} + e^{-Jx} e^{x} + e^{-J} e^{2x} + \ldots + e^{Jx} =$

$$= e^{-Jx} \underbrace{\left(1 + e^{x} + e^{2x} + \ldots + e^{2Jx}\right)}_{\text{geometric series}}$$

$$\frac{1 - e^{(2J+1)x}}{1 - e^{x}}$$

$$Z = e^{-Jx} \frac{1 - e^{(2J+1)x}}{e^{x/2}\left(e^{-x/2} - e^{x/2}\right)} = \frac{e^{-(J+\frac{1}{2})x} - e^{(J+\frac{1}{2})x}}{e^{-\frac{x}{2}} - e^{\frac{x}{2}}} =$$

$$= \frac{\operatorname{sh}\left[(2J+1)\frac{x}{2}\right]}{\operatorname{sh}\left(\frac{x}{2}\right)}$$

$$\frac{\partial \ln Z}{\partial x} = \frac{\operatorname{ch}(J+\frac{1}{2})x}{\operatorname{sh}(J+\frac{1}{2})x} \left(J+\frac{1}{2}\right) - \frac{\operatorname{ch}\frac{x}{2}}{\operatorname{sh}\frac{x}{2}} \cdot \frac{1}{2}$$

$$\Rightarrow M = \left(n g_J \mu_B J\right) \left\{ \frac{2J+1}{2J} \coth\left(\frac{2J+1}{2J} \frac{\mu_B g_J B J}{k_B T}\right) - \frac{1}{2J} \coth\left(\frac{1}{2J} \frac{\mu_B g_J B J}{k_B T}\right) \right\} . \equiv M_{sat} \, B_J\left(\frac{\mu_B g_J B J}{k_B T}\right)$$

Brillouin function

Saturation magnetization

Classical limit: $J \to \infty$, $\mu_B g_J J B = \mu \cdot B$

$$M = M_{sat} \cdot \underbrace{\left(\coth\left(\frac{\mu B}{k_B T}\right) - \frac{k_B T}{\mu B} \right)}_{\text{Langevin function}}$$

$J = \frac{1}{2}$ limit

$$M = M_s \left\{ 2 \coth\left(\frac{\mu_B g_J B}{k_B T}\right) - \coth\left(\frac{\mu_B g_J B}{2 k_B T}\right) \right\}$$

let $y = \dfrac{g_J \mu_B B J}{k_B T} \to \dfrac{g_J \mu_B B}{2 k_B T}$

$$\Rightarrow \frac{M}{M_{sat}} = \left(2 \coth 2y - \coth y \right)$$

But $2 \coth 2y - \coth y = \dfrac{2(e^{2y} + e^{-2y})}{e^{2y} - e^{-2y}} - \dfrac{e^y + e^{-y}}{e^y - e^{-y}} =$

$$= \frac{2(e^{2y} + e^{-2y}) - (e^y + e^{-y})^2}{(e^y - e^{-y})(e^y + e^{-y})} = \frac{e^{2y} + e^{-2y} - 2}{(e^y - e^{-y})(e^y + e^{-y})} = \frac{(e^y - e^{-y})^2}{(e^y - e^{-y})(e^y + e^{-y})} =$$

$$= \frac{e^y - e^{-y}}{e^y + e^{-y}} = \tanh(y)$$

$$\Rightarrow M = M_{sat} \tanh\left(\frac{g_J \mu_B B}{2 k_B T}\right)$$

Usually $y \ll 1$ at normal temperatures

and $\tanh(y) \simeq y$

$*$ $\Rightarrow$ can derive Curie susceptibility $\chi = \dfrac{n \mu_0 \mu_{eff}^2}{3 k_B T}$

Consider now small-B limit: $\frac{\mu_B g_J J B}{k_B T} \equiv y \ll 1$

$$\frac{M}{M_{sat}} = \frac{2J+1}{2J} \coth\left(\frac{2J+1}{2J} y\right) - \frac{1}{2J} \coth\left(\frac{1}{2J} y\right)$$

But $\coth(x) = \frac{1}{x} + \frac{x}{3} + \dots$ for $x \to 0$

$$\Rightarrow \frac{M}{M_{sat}} = \frac{2J+1}{2J}\left(\frac{2J}{2J+1}\frac{1}{y} + \frac{2J+1}{2J}\frac{y}{3}\right) -$$

$$- \frac{1}{2J}\left(\frac{2J}{y} + \frac{1}{2J}\frac{y}{3}\right) = \frac{1}{y} - \frac{1}{y} +$$

$$+ \left(\left(\frac{2J+1}{2J}\right)^2 - \frac{1}{(2J)^2}\right)\frac{y}{3} = \frac{4J+1+4J^2-1}{4J^2}\frac{y}{3} =$$

$$= \frac{J+1}{3J} y$$

$$M = \underbrace{\left(n \mu_B g_J J\right)}_{M_{sat}} \frac{J+1}{3J} \underbrace{\frac{\mu_B g_J J B}{k_B T}}_{y}$$

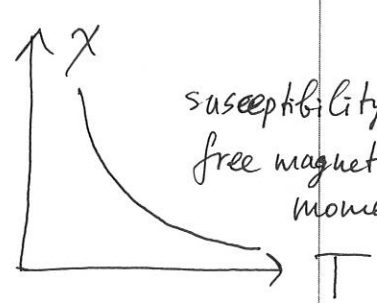
Susceptibility $\chi = \frac{M}{H} \approx \frac{\mu_0 M}{B} = \frac{n \mu_0 \mu_B g_J J(J+1)}{3 k_B T} =$

$$= \frac{n \mu_0 \mu_{eff}^2}{3 k_B T}, \quad \mu_{eff} = \mu_B g_J \sqrt{J(J+1)}$$

effective magnetic moment.

This is <u>Curie law</u>: $\chi = \frac{C_{curie}}{T}$

$$C_{curie} = \frac{n \mu_0 \mu_{eff}^2}{3 k_B}$$



susceptibility of a free magnetic moment.